

DNS 512³ HIT Turbulence Analysis

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This report presents a turbulence analysis of a Direct Numerical Simulation (DNS) of Homogeneous Isotropic Turbulence (HIT) on a **512³ grid** with **1,000,000 time steps**. The simulation operates at a Mach number of **Ma = 0.0164** (essentially incompressible) and a Knudsen number of **Kn = 0.00381**, well within the continuum regime. Key parameters include a kinematic viscosity of **$\nu = 0.00220$** lattice units and a reference velocity of **$u_0 = 0.043$** lattice units.

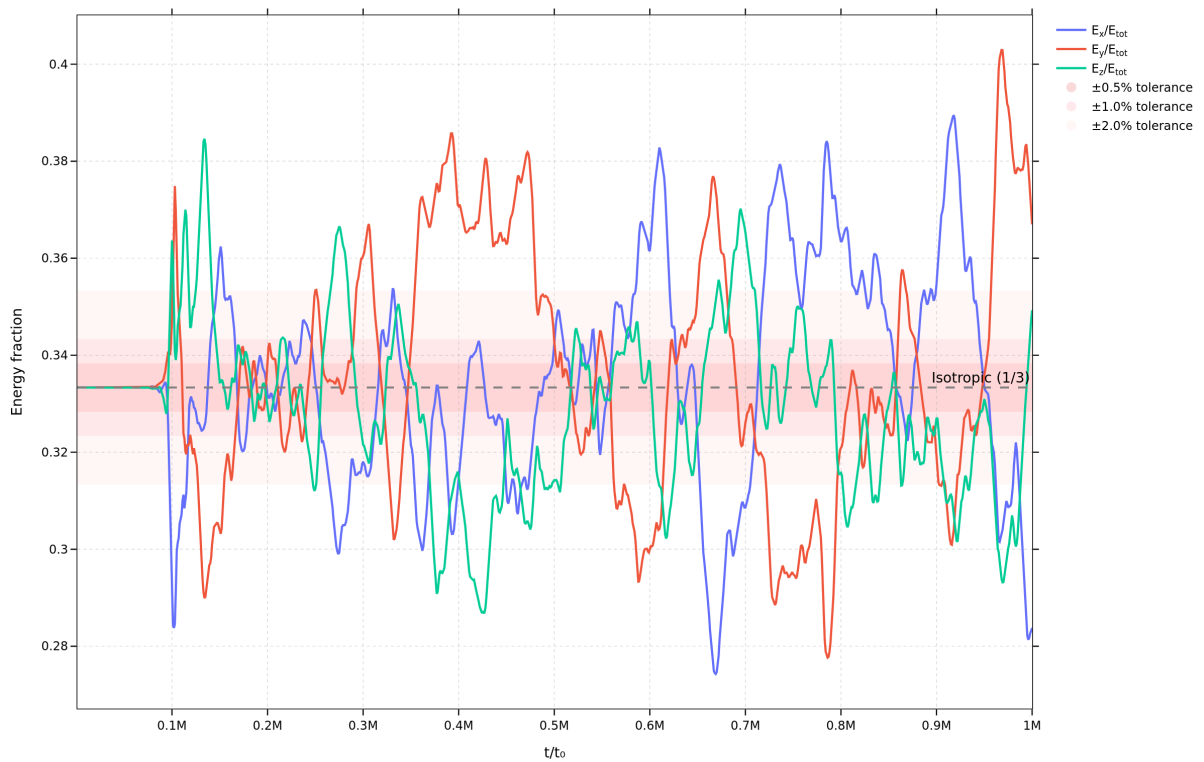


Figure 2. Figure 1: Real-space isotropy analysis showing energy fractions E_x , E_y , E_z relative to total kinetic energy over normalized time t/t_0 . The dashed line at $1/3$ marks the isotropic reference. Colored tolerance bands indicate $\pm 0.5\%$, $\pm 1\%$, and $\pm 2\%$ deviation levels. All three components converge close to the $1/3$ isotropic value, confirming that the DNS achieves strong real-space isotropy.

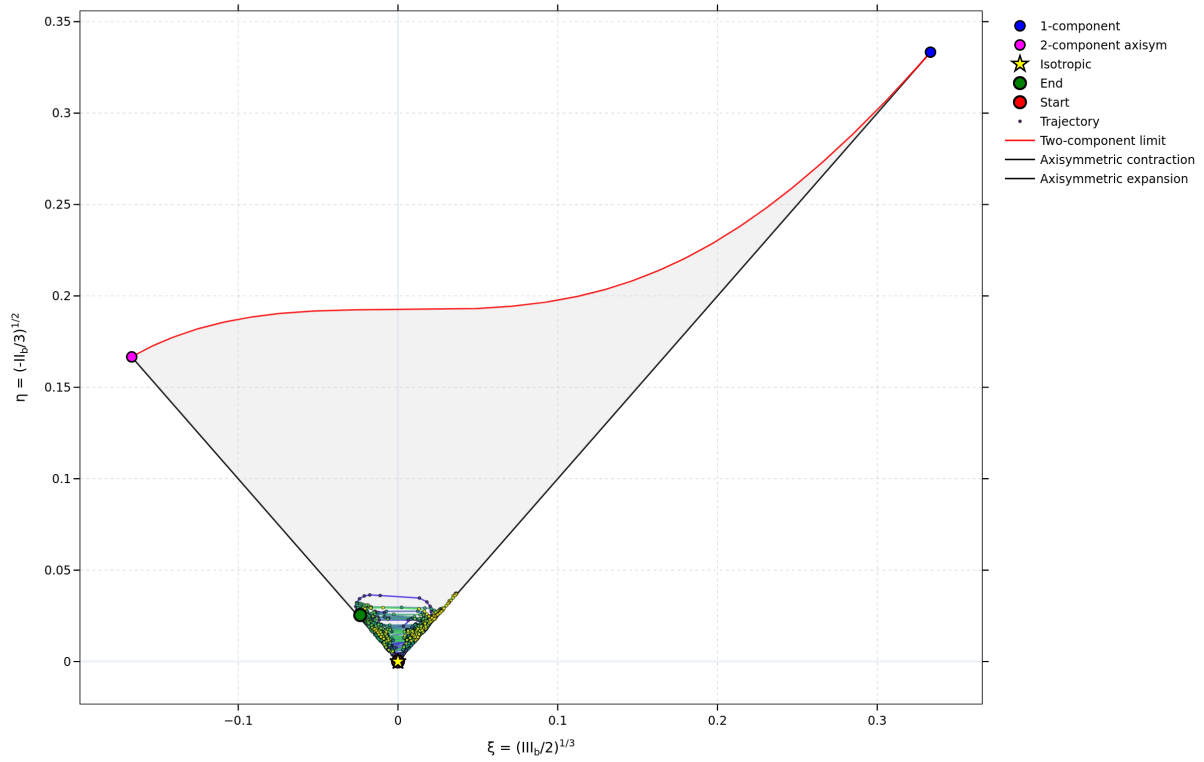


Figure 3. Figure 2: Lumley triangle (ξ, η) showing the trajectory of the Reynolds stress anisotropy tensor invariants. The curvilinear boundaries define the physically realizable domain: axisymmetric expansion, axisymmetric contraction, the two-component limit, and the one-component limit. The DNS trajectory (color-coded by time, starting at red and ending at green) approaches the isotropic origin ($\xi=0, \eta=0$, yellow star), demonstrating convergence toward the isotropic state.